Charotar University of Science and Technology

M.Sc. (Nano Science & Technology) Semester I Examination Dec. 2009

NT701 Concepts of Physical Sciences I

Date: 01 December 2009 **Time: 10.30 am to 13.00 pm** **Max. Marks 50**

Part B

(Write each section on separate answer book)

Section I

Que 1. (a) Describe the seven crystal systems with suitable examples. (3)

(b) Why X-rays are used in crystallographic diffraction? State Bragg's condition of XRD. (3)

(c) An FCC crystal has an atomic radius of 1.246 Å. What are the d_{200}, d_{220} and d_{111}spacings? (4)

OR

(a) Draw the planes of a cubic crystal with proper axis having Miller indices of [111], [110], and [002]. (3)

(b) Find the angle between the two planes [1,1,1] and [1,-1,1]. (3)

(c) The Bragg angle corresponding to the first order reflection from [1,1,1] planes in a crystal is 30° when X-rays of wavelength 1.75 Å are used. Calculate the inter-atomic spacing. (4)

Que 2. (a) Define Meissner effect. (2)

(b) Write three differences between ferromagnetic and anti-ferromagnetic substances. (3)

(c) Why magnetic domains forms in ferromagnetic substances? Explain in brief with diagram. (3)

(d) Discuss energy gap in a superconductor (2)

Section II

Que 1. (a) State the fundamental postulates of quantum mechanics. Give time independent Schrödinger equation and write its solution. Describe the physical significance of the wave function. (6)

(b) Find the lowest energy level and the momentum of an electron in one-dimensional potential well of width 1 Å. (4)

OR

(a) Write an expression for the energy levels of a particle enclosed within an infinite potential well and determine the energy values of an electron confined in a box of width 1 Å. (4)

(b) A particle is moving in one-dimensional potential box (of infinite height) of width 50 Å. Calculate the probability of finding the particle within an interval of 10 Å at the centre of the box when it is in its state of least energy. (4)

(c) Find the lowest energy of an electron confined to move in a cubical box of length 0.5 Å. (2)

Que 2. (a) State the concept and importance of phase space. (3)

(b) What do you mean by ensembles? Distinguish between micro-canonical, canonical and grand canonical ensemble. Give suitable examples of each. (4)

(c) The Fermi energy of silver is 5.51 eV. What is the average energy of a free electron at 0 K ? (3)

Que 3. (a) What are donor levels? (2)

(b) How will you determine the band gap of an intrinsic semiconductor? (5)

(c) The resistivity of intrinsic germanium at 300 K is 0.47 Ωm. If the electron and hole mobility's are 0.38 and 0.18 $m^2V^{-1}s^{-1}$, calculate the intrinsic carrier density at 300 K. (3)

OR

(a) How does the conductivity of an extrinsic semiconductor vary with temperature? (3)

(b) The following data are given for an intrinsic semiconductor at 27 °C:
$n_i = 2.4 \times 1019\ m^{-3}$, $\mu_e = 0.39\ m^2V^{-1}s^{-1}$ and $\mu_p = 0.19\ m^2V^{-1}s^{-1}$
Calculate the conductivity of the intrinsic semiconductor. (3)

(c) Find the resistance at 300 K of an intrinsic Ge rod which is 1 cm long, 1 cm wide and 1 cm thick. The intrinsic carrier density at 300 K is $2.5 \times 10^{19}\ m^{-3}$ and the mobility's of electrons and holes are 0.39 and 0.19 $m^2V^{-1}s^{-1}$ respectively. (4)

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